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Inventor(s)	Wise; Austin et al.

End of life indicators for robotic surgical instruments

Abstract

A surgical tool includes a drive housing defining an indicator aperture, a drive input rotatably coupled to a bottom of the drive housing, and an indicator assembly arranged within the drive housing and actuatable to provide visual indication that the surgical tool has exhausted its useful life. The indicator assembly includes a capstan assembly operatively coupled to the drive input such that rotation of the drive input correspondingly rotates the capstan assembly, an indicator mount secured to the drive housing, and an indicator shaft extending from the indicator mount along a longitudinal axis coaxially aligned with the indicator aperture. The capstan assembly is rotated with the drive input to actuate the indicator assembly between a non-activated state, where the indicator shaft is recessed into the indicator aperture, and an activated state, where the indicator shaft extends out of the drive housing via the indicator aperture to provide the visual indication.

Inventors: **Wise; Austin (Cincinnati, OH), Hibner; John A. (Mason, OH), Inoue; Shinnosuke (Daly City, CA), Remm; Tom (Cincinnati, OH)**

Applicant: **Cilag GmbH International** (Zug, CH)

Family ID: **90730460**

Assignee: **Cilag GmbH International (Zug, CH)**

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Primary Examiner: Chukwurah; Nathaniel C

Attorney, Agent or Firm: Vorys, Sater, Seymour and Pease LLP

Background/Summary

BACKGROUND

(1) Minimally invasive surgical (MIS) instruments are often preferred over traditional open surgical devices due to reduced post-operative recovery time and minimal scarring. During MIS procedures, a variety of instruments and surgical tools may be introduced into the abdominal cavity to engage and/or treat tissue in a number of ways to achieve a diagnostic or therapeutic effect. Various robotic systems have recently been developed to assist in MIS procedures by controlling such MIS instruments. A user (e.g., a surgeon) is able to remotely operate an MIS instrument's end effector by grasping and manipulating in space one or more controllers of the robotic system that communicate with a tool driver coupled to the surgical instrument. User inputs are processed by a computer system incorporated into the robotic surgical system and the tool driver responds by actuating the cable driven motion system and, more particularly, the drive cables. Moving the drive cables articulates the end effector to desired positions and configurations.

(2) MIS instruments have limited life spans. For example, some MIS instruments are designed to expire after a predetermined number of uses or after a set period of time. In some cases, MIS instruments may include an indicator that provides indication when the useful life of the MIS

instruments has been exhausted. Conventional instrument indicators are mechanically powered by one of the MIS instrument's tool drivers, which necessarily decreases overall functionality of the MIS instrument as such tool driver could instead be utilized for other tool functions. Moreover, conventional instrument indicators are not easily recognized and, consequently, sterilization workers often do not notice expired MIS instruments and are accidentally cleaned, sterilized, stored, and later sent to the operating room, despite having no useful operational life remaining. Once discovered in the operating room, personnel will be required to discard the MIS instrument and obtain a replacement. This results in frustration, procedural delay, and possible additional sedation time for the patient. Thus, it may be beneficial to provide indicators that do not utilize tool drivers and indicators that are more easily recognized.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following figures are included to illustrate certain aspects of the present disclosure, and should not be viewed as exclusive embodiments. The subject matter disclosed is capable of considerable modifications, alterations, combinations, and equivalents in form and function, without departing from the scope of this disclosure.

(2) FIG. **1** is a block diagram of an example robotic surgical system that may incorporate some or all of the principles of the present disclosure.

(3) FIG. **2** is an isometric side view of an example surgical tool that may incorporate some or all of the principles of the present disclosure.

(4) FIG. **3** illustrates potential degrees of freedom in which the wrist of the surgical tool of FIG. **2** may be able to articulate (pivot) and translate.

(5) FIG. **4** is a bottom view of the drive housing of FIG. **2**, according to one or more embodiments.

(6) FIGS. **5**A and **5**B are enlarged isometric views of the drive housing of FIG. **2**, according to one or more embodiments.

(7) FIG. **6** is an enlarged, exposed view of the interior of the drive housing, according to one or more embodiments.

(8) FIG. **7**A is an isometric view of the indicator assembly **220** of FIG. **6**, according to one or more embodiments.

(9) FIG. **7**B is an enlarged view of the indicator assembly of FIG. **7**A illustrating an example actuation pathway, according to one or more embodiments.

(10) FIG. **7**C is another enlarged, isometric view of the indicator assembly of FIG. **7**A, according to one or more embodiments.

(11) FIG. **8**A is an enlarged, isometric view of another example indicator assembly, according to one or more additional embodiments.

(12) FIG. **8**B is another enlarged, isometric view of the indicator assembly of FIG. **8**A, according to one or more additional embodiments.

(13) FIG. **9** is an enlarged, isometric view of the indicator assembly of FIGS. **8**A-**8**B, in conjunction with a secondary function, according to one or more embodiments.

(14) FIGS. **10**A-**10**B are enlarged, isometric assembled and exploded views, respectively, of another example indicator assembly, according to one or more additional embodiments.

DETAILED DESCRIPTION

(15) The present disclosure is related to robotic surgical systems and, more particularly, to tool life indicators for surgical tools.

(16) Embodiments discussed herein describe a surgical tool that provides a visual indicator to a user, operator, or technician when the useful life of the surgical tool has been exhausted. The surgical tool can include a drive housing defining an indicator aperture, a drive input rotatably

coupled to a bottom of the drive housing, and an indicator assembly arranged within the drive housing and actuatable to provide visual indication that the surgical tool has exhausted its useful life. The indicator assembly may include a capstan assembly operatively coupled to the drive input such that rotation of the drive input correspondingly rotates the capstan assembly, an indicator mount secured to the drive housing, and an indicator shaft extending from the indicator mount along a longitudinal axis coaxially aligned with the indicator aperture. The capstan assembly may be rotated with the drive input to actuate the indicator assembly between a non-activated state, where the indicator shaft is recessed into the indicator aperture, and an activated state, where the indicator shaft extends out of the drive housing via the indicator aperture to provide the visual indication

(17) FIG. **1** is a block diagram of an example robotic surgical system **100** that may incorporate some or all of the principles of the present disclosure. As illustrated, the system **100** can include at least one set of user input controllers **102***a* and at least one control computer **104**. The control computer **104** may be mechanically and/or electrically coupled to a robotic manipulator and, more particularly, to one or more robotic arms **106** (alternately referred to as “tool drivers”). In some embodiments, the robotic manipulator may be included in or otherwise mounted to an arm cart capable of making the system portable. Each robotic arm **106** may include and otherwise provide a location for mounting one or more surgical instruments or tools **108** for performing various surgical tasks on a patient **110**. Operation of the robotic arms **106** and associated tools **108** may be directed by a clinician **112***a* (e.g., a surgeon) from the user input controller **102***a*.

(18) In some embodiments, a second set of user input controllers **102***b* (shown in dashed line) may be operated by a second clinician **112***b* to direct operation of the robotic arms **106** and tools **108** via the control computer **104** and in conjunction with the first clinician **112***a*. In such embodiments, for example, each clinician **112***a,b* may control different robotic arms **106** or, in some cases, complete control of the robotic arms **106** may be passed between the clinicians **112***a,b* as needed. In some embodiments, additional robotic manipulators having additional robotic arms may be utilized during surgery on the patient **110**, and these additional robotic arms may be controlled by one or more of the user input controllers **102***a,b*.

(19) The control computer **104** and the user input controllers **102***a,b* may be in communication with one another via a communications link **114**, which may be any type of wired or wireless telecommunications means configured to carry a variety of communication signals (e.g., electrical, optical, infrared, etc.) according to any communications protocol. In some applications, for example, there is a tower with ancillary equipment and processing cores designed to drive the robotic arms **106**.

(20) The user input controllers **102***a,b* generally include one or more physical controllers that can be grasped by the clinicians **112***a,b* and manipulated in space while the surgeon views the procedure via a stereo display. The physical controllers generally comprise manual input devices movable in multiple degrees of freedom, and which often include an actuatable handle for actuating the surgical tool(s) **108**, for example, for opening and closing opposing jaws, applying an electrical potential (current) to an electrode, or the like. The control computer **104** can also include an optional feedback meter viewable by the clinicians **112***a,b* via a display to provide a visual indication of various surgical instrument metrics, such as the amount of force being applied to the surgical instrument (i.e., a cutting instrument or dynamic clamping member).

(21) FIG. **2** is an isometric side view of an example surgical tool **200** that may incorporate some or all of the principles of the present disclosure. The surgical tool **200** may be the same as or similar to the surgical tool(s) **108** of FIG. **1** and, therefore, may be used in conjunction with a robotic surgical system, such as the robotic surgical system **100** of FIG. **1**. Accordingly, the surgical tool **200** may be designed to be releasably coupled to a tool driver included in the robotic surgical system **100**. In other embodiments, however, aspects of the surgical tool **200** may be adapted for use in a manual or hand-operated manner, without departing from the scope of the disclosure.

(22) As illustrated, the surgical tool **200** includes an elongated shaft **202**, an end effector **204**, a wrist **206** (alternately referred to as a "wrist joint" or an "articulable wrist joint") that couples the end effector **204** to the distal end of the shaft **202**, and a drive housing **208** coupled to the proximal end of the shaft **202**. In applications where the surgical tool is used in conjunction with a robotic surgical system (e.g., the robotic surgical system **100** of FIG. **1**), the drive housing **208** can include coupling features that releasably couple the surgical tool **200** to the robotic surgical system.
(23) The terms "proximal" and "distal" are defined herein relative to a robotic surgical system having an interface configured to mechanically and electrically couple the surgical tool **200** (e.g., the housing **208**) to a robotic manipulator. The term "proximal" refers to the position of an element closer to the robotic manipulator and the term "distal" refers to the position of an element closer to the end effector **204** and thus further away from the robotic manipulator. Alternatively, in manual or hand-operated applications, the terms "proximal" and "distal" are defined herein relative to a user, such as a surgeon or clinician. The term "proximal" refers to the position of an element closer to the user and the term "distal" refers to the position of an element closer to the end effector **204** and thus further away from the user. Moreover, the use of directional terms such as above, below, upper, lower, upward, downward, left, right, and the like are used in relation to the illustrative embodiments as they are depicted in the figures, the upward or upper direction being toward the top of the corresponding figure and the downward or lower direction being toward the bottom of the corresponding figure.
(24) During use of the surgical tool **200**, the end effector **204** is configured to move (pivot) relative to the shaft **202** at the wrist **206** to position the end effector **204** at desired orientations and locations relative to a surgical site. To accomplish this, the housing **208** includes (contains) various drive inputs and mechanisms (e.g., gears, actuators, etc.) designed to control operation of various features associated with the end effector **204** (e.g., clamping, firing, cutting, rotation, articulation, etc.). In at least some embodiments, the shaft **202**, and hence the end effector **204** coupled thereto, is configured to rotate about a longitudinal axis A.sub.1 of the shaft **202**. In such embodiments, at least one of the drive inputs included in the housing **208** is configured to control rotational movement of the shaft **202** about the longitudinal axis A.sub.1.
(25) The shaft **202** is an elongate member extending distally from the housing **208** and has at least one lumen extending therethrough along its axial length. In some embodiments, the shaft **202** may be fixed to the housing **208**, but could alternatively be rotatably mounted to the housing **208** to allow the shaft **202** to rotate about the longitudinal axis A.sub.1. In yet other embodiments, the shaft **202** may be releasably coupled to the housing **208**, which may allow a single housing **208** to be adaptable to various shafts having different end effectors.
(26) The end effector **204** can exhibit a variety of sizes, shapes, and configurations. In the illustrated embodiment, the end effector **204** comprises a combination tissue grasper and vessel sealer that include opposing first (upper) and second (lower) jaws **210**, **212** configured to move (articulate) between open and closed positions. As will be appreciated, however, the opposing jaws **210**, **212** may alternatively form part of other types of end effectors such as, but not limited to, a surgical scissors, a clip applier, a needle driver, a babcock including a pair of opposed grasping jaws, bipolar jaws (e.g., bipolar Maryland grasper, forceps, a fenestrated grasper, etc.), etc. One or both of the jaws **210**, **212** may be configured to pivot to articulate the end effector **204** between the open and closed positions. In other embodiments, the end effector **204** may not include opposing jaws, but may instead comprise other types of surgical end effectors such as a stapler, a cauterizer, a suction tool, an irrigation tool, and the like.
(27) FIG. **3** illustrates the potential degrees of freedom in which the wrist **206** may be able to articulate (pivot) and thereby move the end effector **204**. The wrist **206** can have any of a variety of configurations. In general, the wrist **206** comprises a joint configured to allow pivoting movement of the end effector **204** relative to the shaft **202**. The degrees of freedom of the wrist **206** are represented by three translational variables (i.e., surge, heave, and sway), and by three rotational

variables (i.e., Euler angles or roll, pitch, and yaw). The translational and rotational variables describe the position and orientation of the end effector **204** with respect to a given reference Cartesian frame. As depicted in FIG. **3**, “surge” refers to forward and backward translational movement, “heave” refers to translational movement up and down, and “sway” refers to translational movement left and right. With regard to the rotational terms, “roll” refers to tilting side to side, “pitch” refers to tilting forward and backward, and “yaw” refers to turning left and right.

(28) The pivoting motion can include pitch movement about a first axis of the wrist **206** (e.g., X-axis), yaw movement about a second axis of the wrist **206** (e.g., Y-axis), and combinations thereof to allow for 360° rotational movement of the end effector **204** about the wrist **206**. In other applications, the pivoting motion can be limited to movement in a single plane, e.g., only pitch movement about the first axis of the wrist **206** or only yaw movement about the second axis of the wrist **206**, such that the end effector **204** moves only in a single plane.

(29) Referring again to FIG. **2**, the surgical tool **200** may also include a plurality of drive cables (obscured in FIG. **2**) that form part of a cable driven motion system configured to facilitate actuation and articulation of the end effector **204** relative to the shaft **202**. Moving (actuating) one or more of the drive cables moves the end effector **204** between an unarticulated position and an articulated position. The end effector **204** is depicted in FIG. **2** in the unarticulated position where a longitudinal axis A.sub.2 of the end effector **204** is substantially aligned with the longitudinal axis A.sub.1 of the shaft **202**, such that the end effector **204** is at a substantially zero angle relative to the shaft **202**. Due to factors such as manufacturing tolerance and precision of measurement devices, the end effector **204** may not be at a precise zero angle relative to the shaft **202** in the unarticulated position, but nevertheless be considered “substantially aligned” thereto. In the articulated position, the longitudinal axes A.sub.1, A.sub.2 would be angularly offset from each other such that the end effector **204** is at a non-zero angle relative to the shaft **202**.

(30) In some embodiments, the surgical tool **200** may be supplied with electrical power (current) via a power cable **214** coupled to the housing **208**. In other embodiments, the power cable **214** may be omitted and electrical power may be supplied to the surgical tool **200** via an internal power source, such as one or more batteries, capacitors, or fuel cells. In such embodiments, the surgical tool **200** may alternatively be characterized and otherwise referred to as an “electrosurgical instrument” capable of providing electrical energy to the end effector **204**.

(31) The power cable **214** may place the surgical tool **200** in electrical communication with a generator **216** that supplies energy, such as electrical energy (e.g., radio frequency energy), ultrasonic energy, microwave energy, heat energy, or any combination thereof, to the surgical tool **200** and, more particularly, to the end effector **204**. Accordingly, the generator **216** may comprise a radio frequency (RF) source, an ultrasonic source, a direct current source, and/or any other suitable type of electrical energy source that may be activated independently or simultaneously.

(32) In applications where the surgical tool **200** is configured for bipolar operation, the power cable **214** will include a supply conductor and a return conductor. Current can be supplied from the generator **216** to an active (or source) electrode located at the end effector **204** via the supply conductor, and current can flow back to the generator **216** via a return electrode located at the end effector **204** via the return conductor. In the case of a bipolar grasper with opposing jaws, for example, the jaws serve as the electrodes where the proximal end of the jaws are isolated from one another and the inner surface of the jaws (i.e., the area of the jaws that grasp tissue) apply the current in a controlled path through the tissue. In applications where the surgical tool **200** is configured for monopolar operation, the generator **216** transmits current through a supply conductor to an active electrode located at the end effector **204**, and current is returned (dissipated) through a return electrode (e.g., a grounding pad) separately coupled to a patient's body.

(33) The surgical tool **200** may further include a manual release switch **218** that may be manually actuated by a user (e.g., a surgeon) to override the cable driven system and thereby manually

articulate or operate the end effector **204**. The release switch **218** is movably positioned on the drive housing **208**, and a user is able to manually move (slide) the release switch **218** from a disengaged position, as shown, to an engaged position. In the disengaged position, the surgical tool **200** is able to operate as normal. As the release switch **218** moves to the engaged position, however, various internal component parts of the drive housing **208** are simultaneously moved, thereby resulting in the jaws **210**, **212** opening, which might prove beneficial for a variety of reasons. In some applications, for example, the release switch **218** may be moved in the event of an electrical disruption that renders the surgical tool **200** inoperable. In such applications, the user would be able to manually open the jaws **210**, **212** and thereby release any grasped tissue and remove the surgical tool **200**. In other applications, the release switch **218** may be actuated (enabled) to open the jaws **210**, **212** in preparation for cleaning and/or sterilization of the surgical tool **200**.

(34) According to embodiments of the present disclosure, the surgical tool **200** may further include a tool end of life indicator assembly **220** that may be automatically activated (triggered) to provide a visual indication that the useful life of the surgical tool **200** has been exhausted and/or that the recommended lifespan of the surgical tool **200** has expired. The tool of life indicator assembly **220** may be alternately referred to herein as "the indicator assembly **220**". Upon activation of the indicator assembly **220**, the user will be visually notified that the service life of the surgical tool **200** has been exhausted and should not be cleaned for re-use but instead decommissioned (e.g., discarded).

(35) In some examples, the surgical tool **200** may include a single tool end of life indicator assembly **220**. In other examples, the surgical tool **200** may include a plurality of tool end of life indicator assemblies **220**, where a first is activated after a first use, a second is activated after a second use, and so on; and activation of all of the plurality of tool end of life indicator assemblies **220** indicates that the surgical tool **200** has reached the end of its life. In at least one embodiment, the tool end of life indicator assembly **220** may provide a visual indication that the surgical tool **200** has a certain amount of life (or uses or hours of use) remaining.

(36) Various metrics may be implemented to measure the useful life of the surgical tool **200**. For example, the "useful life" may be determined by the number of procedures that the surgical tool **200** has been utilized (e.g., twenty procedures). In such embodiments, once the number of uses of the surgical tool **200** reaches a predetermined threshold, the indicator assembly **220** may be activated to visually inform the user. Alternatively, the "useful life" may be determined by the number of hours that the surgical tool **200** has been utilized, the number of articulations or movements that the surgical tool **200** has made, or any combination thereof. The indicator assembly **220** may provide a visually perceivable indication that the surgical tool **200** has exhausted its useful life or is expired, and/or that the surgical tool **200** has a certain amount of life (e.g., uses, hours of use, etc.) remaining.

(37) As described in more detail below, the tool end of life indicator assembly **220** may include a mechanically actuated indicator button or shaft **222** that becomes visible (exposed) once the useful life of the surgical tool **200** has been exhausted or reached. In one or more embodiments, as illustrated, the indicator assembly **220** may be located on the drive housing **208**, such as on a top surface of the drive housing **208**. The indicator assembly **220**, however, may be located at any location on the surgical tool **200** that sufficiently enables a user to visually notice the indicator shaft **222**. During normal operation of the surgical tool **200**, and before reaching the predetermined useful life threshold, the indicator shaft **222** may be recessed into the interior of the drive housing **208** and otherwise not visible. Once it is determined that the useful life of the surgical tool **200** has been exhausted, however, the indicator assembly **220** may be actuated (activated), which results in the indicator shaft **222** extending (protruding) a short distance out of the drive housing **208** to provide a visual indication to the user. The indicator assembly **220** is shown in FIG. **2** with the indicator shaft **222** in the activated state.

(38) FIG. **4** is a bottom view of the drive housing **208**, according to one or more embodiments. As illustrated, the drive housing **208** may include a tool mounting portion **402** used to operatively couple the drive housing **208** to a tool driver of a robotic manipulator. The tool mounting portion **402** may releasably couple the drive housing **208** to a tool driver in a variety of ways, such as by clamping thereto, clipping thereto, or slidably mating therewith. In some embodiments, the tool mounting portion **402** may include an array of electrical connecting pins, which may be coupled to an electrical connection on the mounting surface of the tool driver. While the tool mounting portion **402** is described herein with reference to mechanical, electrical, and magnetic coupling elements, it should be understood that a wide variety of telemetry modalities might be used, including infrared, inductive coupling, or the like.
(39) The tool mounting portion **402** includes and otherwise provides an interface **404** configured to mechanically, magnetically, and/or electrically couple the drive housing **208** to the tool driver. As illustrated, the interface **404** includes and supports a plurality of drive inputs, shown as drive inputs **406***a*, **406***b*, **406***c*, **406***d*, **406***e*, and **406***f*. Each drive input **406***a-f* comprises a rotatable disc configured to align with and couple to a corresponding actuator or “drive output” of a tool driver, such that rotation (actuation) of a given drive output drives (rotates) a corresponding one of the drive inputs **406***a-f*. Each drive input **406***a-f* may provide or define one or more surface features **508** configured to align with mating surface features provided on the corresponding drive output. The surface features **508** can include, for example, various protrusions and/or indentations that facilitate a mating engagement. In some embodiments, some or all of the drive inputs **406***a-f* may include one surface feature **508** that is positioned closer to an axis of rotation of the associated drive input **406***a-f* than the other surface feature(s) **508**. This may help to ensure positive angular alignment of each drive input **406***a-f*.
(40) Actuation of the first drive input **406***a* may be configured to control actuation of the tool end of life indicator assembly **220** (FIG. **2**). In some embodiments, actuating the first drive input **406***a* may not only control actuation of the indicator assembly **220**, as described below, but may also control actuation of another feature or operation for the surgical tool **200** (FIG. **2**). In other embodiments, however, actuating the first drive input **406***a* may solely control actuation of the indicator assembly **220**, without departing from the scope of the disclosure.
(41) Actuation of the second drive input **406***b* may be configured to control rotation of the shaft **202** about its longitudinal axis A.sub.1. The shaft **202** may be rotated clockwise or counter-clockwise depending on the rotational actuation of the second drive input **406***b*. In some embodiments, actuation of the second, third, fourth, and fifth drive inputs **406***b-e* may be configured to operate movement (axial translation) of drive cables that form part of a cable driven motion system, which results in the actuation of the wrist **106** (FIG. **4**) and/or articulation (operation) of the end effector **204** (FIG. **4**). In some embodiments, actuation of the sixth drive input **406***f* may be configured to advance and retract a drive rod, and thereby correspondingly advance or retract a knife at the end effector **204**. Each of the drive inputs **406***a-f* may be actuated based on user inputs communicated to the tool driver coupled to the interface **404**, and the user inputs may be received via a computer system incorporated into the robotic surgical system.
(42) The drive housing **208** may house electronics that store unique identification data for the surgical tool **200** (FIG. **2**). Upon mounting the drive housing **208** to a tool driver of the robotic surgical system **100** (FIG. **1**), the system **100** may be able to identify the type of tool and/or the specific tool utilized in a particular operation based on the unique identification data. In addition, the electronics of the surgical tool **200** may store the useful life of the surgical tool **200** (e.g., the use count), and the useful life of the surgical tool **200** may be determined by logic stored on one or more components of the robotic surgical system **100**. Moreover, the surgical system **100** may store information related to a particular surgical tool **200**, and then access and utilize that stored information when it recognizes that the particular surgical tool **200** is being utilized. For example, the robotic surgical system **100** may recognize that the surgical tool **200** has been installed in the

robotic manipulator and then access its remaining useful life that was previously calculated, so that such useful tool life may be updated as needed following the particular operation in which the surgical tool **200** is being utilized.

(43) In some embodiments, the surgical tool **200** (FIG. **2**) may wirelessly communicate with the robotic surgical system **100** (FIG. **1**). In particular, the robotic surgical system **100** may utilize NFC protocols to identify or authenticate the surgical tool **200** or to associate the surgical tool **200** with stored data related to the surgical tool **200**. In at least some embodiments, the surgical tool **200** includes a tag that may be read remotely and wirelessly, without physical contact, when excited with energy emitted from the robotic manipulator. The tag includes an integrated circuit (or chip) that stores and processes information and modulates and demodulates signals (i.e., radio frequency or "RF" signals) and an antenna that receives and transmits the signal. The tag may include a battery and periodically self-activate to transmit a signal, or may include a battery but activate to transmit a signal when in the presence of the robotic manipulator (or other reader device), or may not include a battery and activate to send a signal when excited by the robotic manipulator (or other reader device). The tag may be read-only, having information assigned thereon, or may be read/write, where information may be written into the tag one or more times. In these examples, the robotic manipulator (or reader device) transmits an encoded radio signal to interrogate the tag within the surgical tool **200**. The tag receives the encoded radio signal and responds by sending the identification and/or other information stored in the integrated circuit (e.g., serial number, use count, usage time, manufacture date, expiration date, etc.) to the robotic manipulator so that it may be analyzed by the robotic surgical system **100**. Accordingly, the robotic surgical system **100** may be able to differentiate between a variety of surgical tools as the tags of each surgical tool will include unique identification data.

(44) FIGS. **5**A and **5**B are enlarged isometric views of the drive housing **208**, according to one or more embodiments. FIGS. **5**A-**5**B also depict example actuation of the tool end of life indicator assembly **220**, which includes the indicator shaft **222**. In FIG. **5**A, the indicator assembly **220** is in a first or "non-activated" state, and thus the indicator shaft **222** is not readily perceivable. However, in FIG. **5**B, the indicator assembly **220** is transitioned to a second or "activated" state, and the indicator shaft **222** extends out of the drive housing **208** a short distance and is readily perceivable by an operator. The remaining portions of the indicator assembly **220** are housed within the drive housing **208**, and various embodiments of the indicator assembly **220** will be discussed below.

(45) When the indicator assembly **220** is in the non-activated state, as shown in FIG. **5**A, the indicator shaft **222** is substantially or entirely received within the drive housing **208**, such that the indicator shaft **222** is mostly or fully occluded from the view of an operator or user. In contrast, when the indicator assembly **220** is transitioned to the activated state, as shown in FIG. **5**B, the indicator shaft **222** is moved such that the top or upper end of the indicator shaft **222** protrudes (extends) a short distance out of the drive housing **208** via an indicator aperture **502** defined in the drive housing **208**. When the indicator shaft **222** protrudes out of the drive housing **208** via the indicator aperture **502**, this provides a positive, visual indication that the surgical tool **200** (FIG. **2**) has exhausted its useful life, and appropriate action should be taken.

(46) In some embodiments, the upper end or top of the indicator shaft **222** may be painted or otherwise include or exhibit a bright color (e.g., red), which may be easily perceivable by a user when transitioned to the activated state. In at least one embodiment, the indicator shaft **222** may include a light (e.g., a light-emitting diode or "LED") that is triggered to emit light (shine) when the indicator assembly **220** transitions to the activated state. In some embodiments, the indicator assembly **220** may be configured and otherwise designed such the indicator shaft **222** cannot be advanced back into the drive housing **208** without disassembling the entire drive housing **208**.

(47) FIG. **6** is an enlarged, exposed view of a portion of the interior of the drive housing **208**, according to one or more embodiments. Several component parts that would otherwise be contained within the drive housing **208** are not shown in FIG. **6** to enable discussion of the depicted

component parts. As illustrated, the drive housing **208** houses and otherwise contains the indicator assembly **220**. FIG. **6** depicts one example of the indicator assembly **220**, and other designs or configurations of the indicator assembly **220** will be provided below.

(48) In the illustrated embodiment, the indicator assembly **220** includes a capstan assembly **602** operatively coupled to or extending from the first drive input **406***a* (FIG. **4**) such that actuation of the first drive input **406***a* results in rotation of the capstan assembly **602**. The indicator assembly **220** further includes an indicator mount **604** mounted to the drive housing **208** in a manner that allows the indicator mount **604** to move (translate) up and down along a longitudinal axis B.sub.1 relative to the capstan assembly **602** as the capstan assembly **602** rotates. A follower **606** (partially occluded) extends laterally from the indicator mount **604** and is configured to slidingly engage various surfaces and ramps of the capstan assembly **602**, which urges the indicator mount **604** to move vertically (up and down) along the longitudinal axis B.sub.1.

(49) The indicator shaft **222** may form part of or otherwise extend from the indicator mount **604** such that as the indicator mount **604** translates (moves) along the longitudinal axis B.sub.1, the indicator shaft **222** correspondingly moves in the same axial direction along the longitudinal axis B.sub.1. The indicator mount **604** and the indicator shaft **222** are constrained within the drive housing **208** to movement along the longitudinal axis B.sub.1, which may be coaxially aligned with the indicator aperture **502** (FIGS. **5**A-**5**B) defined in the drive housing **208**. Consequently, as the indicator mount **604** moves along the longitudinal axis B.sub.1, the indicator shaft **222** is simultaneously moved toward or away from the indicator aperture **502**, depending on the operation of the indicator mount **604**.

(50) The indicator assembly **220** may further include a coil spring **608** that extends about a portion of the indicator shaft **222**. The bottom of the coil spring **608** may engage or be mounted to the indicator mount **604**, and the top of the coil spring **608** may engage a static portion (not shown) of the drive housing **208**. Consequently, as the follower **606** slidingly engages the surfaces and the ramps of the capstan assembly **602** and thereby urges the indicator mount **604** to move up and along the longitudinal axis B.sub.1, the coil spring **608** will progressively compress and build spring force. In some embodiments, for example, the top of the coil spring **608** may engage the bottom internal surface of the drive housing **208** at or near the indicator aperture **502** (FIGS. **5**A-**5**B). In other embodiments, however, the top of the coil spring **608** may engage another structure arranged within the drive housing **208**, such as a chassis or another stationary surface mounted within or forming part of the drive housing **208**.

(51) FIG. **7**A is an enlarged, isometric view of the indicator assembly **220** of FIG. **6**, according to one or more embodiments. As illustrated, the capstan assembly **602** either forms part of or extends from the first drive input **406***a*, and is rotatable about a central axis C.sub.1 as the first drive input **406***a* is rotated. Moreover, the capstan assembly **602** may include a plurality of structural levels, shown as a first structural level **702***a*, a second structural level **702***b*, a third structural level **702***c*, and a fourth structural level **702***d*. The structural levels **702***a-d* are axially offset from each other along the central axis C.sub.1, and each structural level **702***a-d* includes a planar surface **704** that is parallel to the planar surfaces **704** of the other structural levels **702***a-d*. During actuation of the indicator assembly **220**, the capstan assembly **602** rotates and the follower **606** slidingly engages the planar surfaces **704**. While four structural levels **702***a-d* are shown, the capstan assembly **602** may include more or less than four.

(52) The indicator assembly **220** may further include a plurality of ramps, shown as a first ramp **706***a*, a second ramp **706***b* (occluded), and third ramp **706***c*. Each ramp **706***a-c* provides a means for the follower **606** to transition vertically between vertically adjacent structural levels **702***a-d* and their corresponding planar surfaces **704**. More specifically, the first ramp **706***a* provides a vertical transition from the first structural level **702***a* to the second structural level **702***b*, the second ramp **706***b* provides a vertical transition from the second structural level **702***b* to the third structural level **702***c*, and the third ramp **706***c* provides a vertical transition from the third structural level **702***c* to

the fourth structural level **702***d*.

(53) As the capstan assembly **602** is rotated in a predetermined and precise rotational sequence, the follower **606** may be able to progressively and sequentially engage each structural level **702***a-d* by engaging and sliding (riding) up each ramp **706***a-c*. As the follower **606** transitions to each level **702***a-d*, the indicator mount **604** and the indicator shaft **222** will correspondingly move vertically along the longitudinal axis B.sub.1. Moreover, as the indicator shaft **222** moves upwards along the longitudinal axis B.sub.1, the coil spring **608** will progressively be compressed against the underside (bottom) of a portion **708** of the drive housing **208** (shown as a dashed line in FIG. **7**A) and thereby build spring force.

(54) FIG. **7**B is an enlarged view of the indicator assembly **220** of FIG. **7**A illustrating an example actuation pathway **710** that the follower **606** (FIG. **7**A) must traverse to properly actuate the indicator assembly **220**, according to one or more embodiments. To fully actuate the indicator assembly **220**, and thereby transition the indicator shaft **222** (FIGS. **7**A and **7**C) to the activated state, the follower **606** must precisely traverse the actuation pathway **710** up each ramp **706***a-c* and to each structural level **706***a-d*. The actuation pathway **710** terminates at a trough or pocket **708** defined by the indicator assembly **220** at the top-most, fourth structural level **706***d*.

(55) The indicator assembly **220** will technically be transitioned to the activated state once the follower **606** reaches the fourth structural level **702***d*, but will not be fully actuated until the follower **606** reaches the pocket **708**. More specifically, once the follower **606** reaches the fourth structural level **702***d*, the top or upper end of the indicator shaft **222** will protrude (extend) a short distance out of the drive housing **208** (FIGS. **5**A-**5**B) to provide a positive, visual indication that the surgical tool **200** (FIG. **2**) has exhausted its useful life. However, the indicator assembly **220** will be fully actuated once the follower **606** traverses the planar surface **704** of the fourth structural level **702***d* and is ultimately received within the pocket **708**. Once the follower **606** is received within the pocket **708**, the capstan assembly **602** will be prevented from further rotation in either angular direction, thereby preventing the indicator assembly **220** from returning to the non-activated state.

(56) The follower **606** will be able to traverse the actuation pathway **710** (shown as a plurality of arrows), and thereby transition the indicator assembly **220** to the activated state, by rotating the capstan assembly **602** in a predetermined rotational sequence comprising a precise combination of clockwise and counter-clockwise rotations. The combination of precise rotational movements required to transition the indicator assembly **220** to the activated state may be sufficiently complex such that inadvertently matching the predetermined rotational sequence could only be accomplished in extremely rare circumstances.

(57) Rotation of the capstan assembly **602** will be dictated by actuation of the first drive input **406***a*. As mentioned above, the drive inputs, including the first drive input **406***a*, are configured to align with and couple to a corresponding actuator or "drive output" of a tool driver, and rotation (actuation) of a given drive output drives (rotates) the corresponding drive input. According to embodiments of the present disclosure, once the predetermined operational threshold of the surgical tool **200** (FIG. **2**) has been reached, the robotic manipulator of the tool driver may be operated to actuate the first drive input **406***a* in the predetermined rotational sequence, and thereby transition the indicator assembly **220** to the activated state. Accordingly, the precise combination of clockwise and counter-clockwise rotations may be known and programmed into the control computer **104** (FIG. **1**) used to operate the robotic manipulator.

(58) In the illustrated embodiment, the predetermined rotational sequence for the actuation pathway **710** includes rotating the first drive input **406***a*, and therefore the capstan assembly **602**, in a first angular direction (e.g., counter-clockwise) for a first predetermined angular magnitude **712***a*. As shown in FIG. **7**B, rotating the capstan assembly **602** in the first angular direction for the first predetermined angular magnitude **712***a* may allow the follower **606** (FIGS. **7**A and **7**C) to traverse the planar surface **704** of the first structural level **702***a*, engage and ride up the first ramp **706***a*, and

traverse at least a portion of the planar surface **704** of the second structural level **702***b*.
(59) After rotating for the first predetermined angular magnitude **712***a*, the predetermined rotational sequence may then include changing angular direction and rotating the first drive input **406***a*, and therefore the capstan assembly **602**, in a second angular direction (e.g., clockwise) for a second predetermined angular magnitude **712***b*. Rotating the capstan assembly **602** in the second angular direction for the second angular magnitude **712***b* may allow the follower **606** to engage and ride up the second ramp **706***b*, and subsequently traverse at least a portion of the planar surface **704** of the third structural level **702***c*.
(60) The predetermined rotational sequence may then include changing angular direction again and rotating first drive input **406***a*, and therefore the capstan assembly **602**, in the first angular direction (e.g., counter-clockwise) for a third predetermined angular magnitude **712***c*. Rotating the capstan assembly **602** in the first angular direction for the third angular magnitude **712***c* may allow the follower **606** to engage and ride up the third ramp **706***c*, and subsequently traverse at least a portion of the planar surface **704** of the third structural level **702***d*. Rotating the capstan assembly **602** in the first angular direction for the third angular magnitude **712***c* may also cause the follower **606** to locate and ultimately be received within the pocket **708**.
(61) Advantageously, the capstan assembly **602** may define or otherwise provide one or more cutouts **714** along the actuation pathway **710**. The cutout(s) **714** may comprise a void or open area defined in the capstan assembly **602** that extends to the first (or "bottom-most") structural level **702***a*. The cutout(s) **714** may be configured to ensure that the capstan assembly **602** is rotated precisely in accordance with the predetermined rotational sequence, otherwise the follower **606** (FIGS. **7**A and **7**C) will encounter the cutout(s) **714**. Upon encountering the cutout(s) **714**, the follower **606** will fall into the cutout(s) **714** and return to the first structural level **702***a* under the spring force released by the coil spring **608** (FIG. **7**A). Upon reaching the first structural level **702***a*, the follower **606** must traverse the actuation pathway **710** from the beginning and otherwise start over.
(62) Importantly, each ramp **706***a-c* is keyed and otherwise located at corresponding predetermined angular locations. Consequently, the capstan assembly **602** must be precisely rotated in accordance with the predetermined rotational sequence or else the follower **606** (FIGS. **7**A and **7**C) will not locate or be able to ride up the ramps **706***a-c*. Instead, the follower **606** may risk encountering the cutout(s) **714**, which returns the follower **606** to the first structural level **702***a* and requires the follower **706** to traverse the actuation pathway **710** from the beginning. Moreover, each ramp **706***a-c* may comprise a living hinge that allows the follower **606** to traverse under the ramp **706***a-c* when the capstan assembly **602** is rotated in one angular direction, but allows the follower **606** to engage and ride up the ramp **706***a-c* when the capstan assembly **602** is rotated in the opposite angular direction. Consequently, the capstan assembly **602** must again be precisely rotated in accordance with the predetermined rotational sequence or else the follower **606** will not probably engage or ride up the ramps **706***a-c*.
(63) FIG. **7**C is another enlarged, isometric view of the indicator assembly **220** of FIG. **7**A, according to one or more embodiments. As illustrated, the follower **606** has successfully traversed the actuation pathway **710** (FIG. **7**B) and is shown received within the pocket **708**. Once the follower **606** is received within the pocket **708**, the top or upper end **716** of the indicator shaft **222** will protrude (extend) a short distance out of the drive housing **208** (FIGS. **5**A-**5**B) to provide a positive, visual indication that the surgical tool **200** (FIG. **2**) has exhausted its useful life. Moreover, since the follower **606** and the indicator mount **604** are stationary and grounded to the drive housing **208** (FIG. **6**), once the follower **606** is received within the pocket **708**, the follower **606** will bind against the sidewalls of the pocket **708** and thereby prevent the capstan assembly **602** from further rotation in either angular direction. As a result, the indicator assembly **220** will be prevented from returning to the non-activated state without entirely disassembling the drive housing **208**.

(64) The ramps **706***a-c*, the cutout(s) **714**, and the predetermined rotational sequence required to successfully traverse the actuation pathway **710** (FIG. **7**B) may each be considered fail-safes that ensure that the indicator assembly **220** is not inadvertently or prematurely transitioned to the activated state. For example, following each surgical operation, the surgical tool **200** (FIG. **2**) must be removed from the robotic manipulator and cleaned, sterilized, and processed before being placed back into active duty. During the cleaning and sterilizing process, an operator (technician) may inadvertently rotate the first drive input **406***a*. Because the first drive input **406***a* must be rotated in the precise combination of clockwise and counter-clockwise rotations in accordance with the predetermined rotational sequence, it is virtually impossible for an operator to inadvertently trigger actuation of the indicator assembly **220** to the activated state.
(65) FIG. **8**A is an enlarged, isometric view of another example indicator assembly **800**, according to one or more additional embodiments. The indicator assembly **800** may be similar in some respects to the indicator assembly **220** of FIGS. **6** and **7**A-**7**C, and therefore may be best understood with reference thereto. In at least one embodiment, the indicator assembly **800** may replace the indicator assembly **220** within the drive housing **208** (FIG. **6**), and therefore may be used in conjunction with the surgical tool **200** (FIG. **2**).
(66) In the illustrated embodiment, the indicator assembly **800** includes a capstan assembly **802** operatively coupled to or extending from the first drive input **406***a* such that actuation of the first drive input **406***a* results in rotation of the capstan assembly **802**. The indicator assembly **800** further includes an indicator mount **804**, which may be rotatably mounted to the drive housing **208** (FIG. **6**) and rotatable about a longitudinal axis D.sub.1.
(67) As illustrated, a spring-loaded arm **806** extends from or otherwise forms part of the indicator mount **804**, and a follower **808** extends from an end of the arm **806** such that the follower **808** moves simultaneously with the arm **806** as the indicator mount **804** rotates about the longitudinal axis D.sub.1. In the illustrated embodiment, the follower **808** may comprise a rod or pin that extends orthogonally from the arm **806**. As described in more detail below, the follower **808** may be configured to be received within a channel defined by the capstan assembly **802**.
(68) The indicator assembly **800** further includes an indicator shaft **810**, similar in some respects to the indicator shaft **222** of FIGS. **6** and **7**A-**7**C, and therefore may be best understood with reference thereto. The indicator shaft **810** may form part of or otherwise extend from the indicator mount **804** such that as the indicator mount **804** rotates about the longitudinal axis D.sub.1, the indicator shaft **810** correspondingly rotates in the same angular direction. The longitudinal axis D.sub.1 may be coaxially aligned with the indicator aperture **502** (FIGS. **5**A-**5**B) defined in the drive housing **208** (FIG. **6**).
(69) The indicator assembly **800** may further include a shaft **812** extending from or otherwise forming an integral part of the indicator mount **804**. The shaft may be rotatably mounted to the drive housing **208** (FIG. **6**), or another static structure secured to the drive housing **208**. In some embodiments, as illustrated, the shaft **812** may be received within a receptacle **814** either forming part of the drive housing **208** or otherwise rigidly secured to the drive housing **208**. A torsion spring **816** may be operatively coupled to the indicator mount **804** and configured to naturally and continuously bias the indicator mount **804** and the interconnected follower **808** toward the central axis C.sub.1 of the capstan assembly **802**. In the illustrated embodiment, the torsion spring **816** extends about the shaft **812**, but could alternatively be mounted in yet other locations related to the indicator mount **804**.
(70) The indicator assembly **800** may further include a coil spring **818** that extends about a portion of the indicator shaft **810** and urges the indicator shaft **810** to move vertically upward along the longitudinal axis D.sub.1 in the direction E. The bottom of the coil spring **818** may engage or be mounted to a portion of the indicator mount **804**, such as a receptacle **820** secured to or forming part of the indicator mount **804**. The top of the coil spring **818** may engage a portion the indicator shaft **810** such that when the indicator shaft **810** is released, as described below, the spring force of

the coil spring **818** will cause the indicator shaft **810** to move upwards in the direction E and along the longitudinal axis D.sub.1.

(71) The indicator shaft **810** may be releasably coupled to a vertically static structure; i.e., a structure that is unable to move vertically and otherwise along the longitudinal axis D.sub.1. In some embodiments, the vertically static structure may comprise the receptacle **820**, but could alternatively include other vertically static structures provided within or forming part of the drive housing **208** (FIG. **6**). In the illustrated embodiment, the indicator shaft **810** provides and otherwise defines a lateral projection **822** extending laterally from the main body of the indicator shaft **810**. In some embodiments, the lateral projection **822** may be releasably coupled to the vertically static structure (e.g., the receptacle **820**) via a bayonet connection, a pin and slot configuration, or the like. In such embodiments, as the indicator mount **804** rotates about the longitudinal axis D.sub.1, as described herein, the lateral projection **822** may be released from the vertically static structure. Once the lateral projection **822** is released from the vertically static structure, the spring force of the coil spring **818** may be released to advance the indicator shaft **810** vertically in the direction E and along the longitudinal axis D.sub.1. As indicated above, the longitudinal axis D.sub.1 may be coaxially aligned with the indicator aperture **502** (FIGS. **5**A-**5**B) of the drive housing **208** (FIG. **6**), thus advancing the indicator shaft **810** in the direction E effectively transitions the indicator assembly **800** to the activated state.

(72) The capstan assembly **802** either forms part of or extends from the first drive input **406***a*, and is rotatable about the central axis C.sub.1 as the first drive input **406***a* is rotated. As illustrated, the capstan assembly **802** may include a plurality of disc keys, shown as a first disc key **824***a*, a second disc key **824***b*, and a third disc key **824***c*. The disc keys **824***a-c* are arranged concentric and coaxially aligned with the central axis C.sub.1 and slidably engageable with vertically adjacent disc keys **824***a-c*. Each disc key **824***a-c* provides and otherwise defines one or more radial shoulders **826** and a slot **828**. To actuate the indicator assembly **800** and thereby transition the indicator assembly **800** to the activated state, the slots **828** in each disc key **824***a-c* must align axially, thereby allowing the follower **808** to enter a channel **830** (see FIG. **8**B) that forms and is otherwise created when the slots **828** axially align.

(73) Axially aligning the slots **828** can be accomplished by rotating the capstan assembly **802** in a predetermined rotational sequence comprising a precise combination of clockwise and counter-clockwise rotations. The combination of precise rotational movements required to transition the indicator assembly **800** to the activated state may be sufficiently complex such that inadvertent matching of the predetermined rotational sequence will occur in extremely rare circumstances.

(74) Rotation of the capstan assembly **802** will be dictated by actuation of the first drive input **406***a*. Once the predetermined operational threshold of the surgical tool **200** (FIG. **2**) has been reached and acknowledged, the robotic manipulator of the tool driver may be programmed to actuate the first drive input **406***a* in the predetermined rotational sequence, and thereby transition the indicator assembly **800** to the activated state.

(75) In the illustrated embodiment, the predetermined rotational sequence may include rotating the first drive input **406***a* in a first angular direction (e.g., counter-clockwise) for a first predetermined angular magnitude. As indicated above, the first disc key **824***a* may be secured to the first drive input **406***a*, thus rotating the first drive input **406***a* in the first angular direction correspondingly rotates the first disc key **824***a* in the same angular direction. The first disc key **824***a* may be rotated in the first angular direction for the first predetermined angular magnitude until opposing radial shoulders **826** defined on the first and second disc keys **824***a,b* engage. In some embodiments, at this point the slots **828** of the first and second disc keys **824***a,b* are aligned axially.

(76) The predetermined rotational sequence may then include changing angular direction by rotating the first drive input **406***a*, and therefore the first and second disc keys **824***a,b*, in a second angular direction (e.g., clockwise) for a second predetermined angular magnitude. The first and second disc keys **824***a,b* may be rotated in the second angular direction for the second

predetermined angular magnitude until opposing radial shoulders **826** defined on the second and third disc keys **824***b,c* engage. Once the opposing radial shoulders **826** defined on the first and second disc keys **824***a,b* engage, further rotation of the first disc key **824***a*, and therefore the second disc key **824***b*, will drive the radial shoulders **826** of the second disc **824***b* to engage with the radial shoulders **826** of the third disc **824***c*. As the direction of rotation changes, friction will keep the third disc key **824***c* in place, while the opposing face of the radial shoulder **826** of the first disc **824***a* can drive the second disc key **824***b* to an axially aligned position with the third disc key **824***b*. In some embodiments, at this point the slots **828** of the first, second, and third disc keys **824***a-c* become aligned axially, thus forming the channel **830** (FIG. **8**B).
(77) The predetermined rotational sequence may then include changing angular direction again and rotating first drive input **406***a*, and therefore the first, second, and third disc keys **824***a-c*, in the first angular direction (e.g., counter-clockwise) for a third predetermined angular magnitude. Rotating the first, second, and third disc keys **824***a-c* in the first angular direction for the third angular magnitude may be configured to align the channel **830** (FIG. **8**B) with the follower **808** and thereby allow the follower **808** to enter the channel **830**.
(78) FIG. **8**B is another enlarged, isometric view of the indicator assembly **800** of FIG. **8**A, according to one or more additional embodiments. In FIG. **8**B, the indicator assembly **800** has been actuated and otherwise transitioned to the activated state. More specifically, as the slots **828** defined in the first, second, and third disc keys **824***a-c* align axially, as described above, the channel **830** is created and otherwise defined by the capstan assembly **802**. Once the capstan assembly **802** is rotated to angularly align the channel **830** with the follower **808**, as described above, the spring force of the torsion spring **816** will naturally urge the indicator mount **804** to rotate about the longitudinal axis D.sub.1 in the direction F. Rotating the indicator mount **804** in the direction F will correspondingly move the arm **806** and the follower **808** in the same direction and thereby inserting (receiving) the follower **808** within the channel **830**.
(79) Rotating the indicator mount **804** about the longitudinal axis D.sub.1 in the direction F will correspondingly and simultaneously rotate the indicator shaft **810** in the same angular direction. As the indicator shaft **810** rotates in the direction F, the indicator shaft **810** will be released from the vertically static structure (e.g., the receptacle **820**). More specifically, in applications where the vertically static structure comprises the receptacle **820**, and the indicator shaft **810** is removably coupled to the receptacle **820**, rotating the indicator shaft **810** about the longitudinal axis D.sub.1 will correspondingly release the lateral projection **822** from the receptacle **820**. Once the lateral projection **822** is released from the vertically static structure, the spring force of the coil spring **818** may be released to advance the indicator shaft **810** vertically in the direction E and along the longitudinal axis D.sub.1, thereby transitioning the indicator assembly **800** to the activated state.
(80) Notably, once the follower **808** is received within the channel **830**, the capstan assembly **802** will be prevented from further rotation in either angular direction. As a result, the indicator assembly **800** will be prevented from returning to the non-activated state without entirely disassembling the drive housing **208** and resetting the indicator assembly **800**.
(81) In some embodiments, instead of the indicator shaft **810** extending vertically in the direction E to transition the indicator assembly **800** to the activated state, it is contemplated herein that actuation of the indicator assembly **800** may drive a variety of other features configured to communicate to the user that the instrument has reached its end of life. In other embodiments, for example, the indicator shaft **810** could be replaced with a visual opening in the drive housing **208** (FIG. **6**) showing a state change triggered by actuation of the indicator mount **804**. Alternatively, a portion of the drive housing **208** could be displaced to indicate end of life of the surgical tool **200** (FIG. **2**).
(82) FIG. **9** is an enlarged, isometric view of the indicator assembly **800** of FIGS. **8**A-**8**B, in conjunction with a secondary function, according to one or more embodiments. More specifically, FIG. **9** shows an example embodiment where the indicator assembly **800** may be used not only to

indicate the termination of the useful life of the surgical tool **200** (FIG. **2**), but may also form part of a secondary operational function of the surgical tool **200**.

(83) In the illustrated embodiment, the capstan assembly **802** includes the first drive input **406***a* and the disc keys **824***a-c*, as generally described above. However, the capstan assembly **802** further includes a drive gear **902**, shown as a worm gear, arranged within the drive housing **208** to engage and drive a corresponding driven gear **904**. As the capstan assembly **802** is actuated, via operation of the first drive input **406***a*, the drive gear **902** may drive the driven gear **904** to undertake one or more functions of the surgical tool **200** (FIG. **2**). In one embodiment, for example, driving the driven gear **904** may cause the shaft **202** to rotate about its longitudinal axis A.sub.1 (FIG. **2**). In other embodiments, however, driving the driven gear **904** may carry out other functions or operations, without departing from the scope of the disclosure.

(84) Accordingly, the indicator assembly **800** may be used in conjunction with an ‘active’ capstan or series of active capstans. In such embodiments, actuation of the indicator assembly **800** to the “activated” state would still require a precise combination of clockwise and counter-clockwise rotations, which would cause unique operation of the surgical tool (FIG. **2**) via interaction between the drive and driven gears **902**, **904**. In at least one embodiment, the robotic system could monitor movements in a clinical setting to guarantee that the indicator assembly **800** would not be activated prematurely and intraoperatively. In any event, because of the precise combination of clockwise and counter-clockwise rotations required to actuate the indicator assembly **800**, the chances of the first drive input **406***a* being inadvertently and prematurely rotated in the precise combination would be sufficiently complex and would only be triggered inadvertently on rare circumstances.

(85) FIGS. **10**A-**10**B are enlarged, isometric assembled and exploded views, respectively, of another example indicator assembly **1000**, according to one or more additional embodiments. The indicator assembly **1000** may be similar in some respects to the indicator assemblies **220** and **800** of FIGS. **6**, **7**A-**7**C, and **8**A-**8**B, and therefore may be best understood with reference thereto. In at least one embodiment, the indicator assembly **1000** may replace the indicator assembly **220** within the drive housing **208** (FIG. **6**), and may thus be used in conjunction with the surgical tool **200** (FIG. **2**).

(86) As illustrated, the indicator assembly **1000** includes a capstan assembly **1002** that extends from and otherwise forms part of the first drive input **406***a*. The capstan assembly **1002** may include a central shaft **1004** having a first or “lower” end **1006***a* and a second or “upper” end **1006***b* opposite the lower end **1006***a*. The lower end **1006***a* may be operatively coupled to or form an integral extension of the first drive input **406***a*, and may extend along the central axis C.sub.1.

(87) A torsion spring **1008** extends about the central shaft **1004** and has a first or “lower” end **1010***a* and a second or “upper” end **1010***b*. The lower end **1010***a* may be operatively coupled to the first drive input **406***a*. In the illustrated embodiment, for example, the lower end **1010***a* may be received within a pocket or slot **1012** defined in the first drive input **406***a*. In contrast, when the indicator assembly **1000** is installed in the drive housing **208** (FIG. **6**), the upper end **1010***b* may be secured to a static portion of the drive housing **208** or another static feature housed within the drive housing **208**. As a result, as the first drive input **406***a* rotates, the torsion spring **1008** is configured to resist rotation of the capstan assembly **1002**.

(88) The capstan assembly **1002** may further include a receptacle **1014** arranged or otherwise provided at the upper end **1006***b* of the central shaft **1004**. As best seen in FIG. **10**B, the receptacle **1014** provides or defines a central aperture **1016**. The receptacle **1014** also provides or defines a release notch **1018** that forms a part of the central aperture **1016**.

(89) The capstan assembly **1002** may also include an indicator mount **1020** sized to be received at least partially within the central aperture **1016**. More specifically, the indicator mount **1020** includes a lower extension **1022** (FIG. **10**B) extendable into the central aperture **1016**. The indicator mount **1020** may also provide a lateral projection **1024**. When the indicator assembly **1000** is installed in the drive housing **208** (FIG. **6**), the lateral projection **1024** may be coupled to a

rotationally static portion of the drive housing **208** or another rotationally static feature provided within the drive housing **208**. The term "rotationally static portion" refers to a portion of the drive housing **208** or other feature within the drive housing **208** that will prevent the indicator mount **1020** from rotating about the central axis C.sub.1, but will allow the indicator mount **1020** to move vertically along the central axis C.sub.1 when the indicator assembly **1000** is actuated to the activated state.
(90) As illustrated, a follower or "release member" **1026** (FIG. **10**B) may be provided and otherwise defined on the outer radial surface of the lower extension **1022**. In some embodiments, the release member **1026** may comprise a radial projection or protrusion extending radially outward from the outer radial surface of the lower extension **1022**. The release member **1026** may be sized to be received by and through the release notch **1018** when the capstan assembly **1002**, and more particularly the central shaft **1004** and the receptacle **1014**, is rotated about the central axis C.sub.1 to align the release notch **1018** with the release member **1026**.
(91) The capstan assembly **1002** further includes an indicator shaft **1028** that extends from and otherwise forms an integral part or extension of the indicator mount **1020**. The indicator shaft **1028** extends along the central axis C.sub.1, which is coaxially aligned with the indicator aperture **502** (FIGS. **5**A-**5**B) of the drive housing **208** (FIG. **6**). Consequently, moving the indicator shaft **1028** upwards along the central axis C.sub.1 will extend the indicator shaft **1028** through the indicator aperture **502** and effectively transition the indicator assembly **1000** to the activated state.
(92) The indicator assembly **1000** may further include a mounting shaft **1030** (FIG. **10**B) extending from or otherwise forming an integral part of the indicator mount **1020**. In the illustrated embodiment, the mounting shaft **1030** extends from the lower extension **1022**, and extends coaxially with the central axis C.sub.1. The mounting shaft **1030** may be sized and otherwise configured to be received within the central aperture **1016**. A coil spring **1032** (FIG. **10**B) may extend about a portion of the mounting shaft **1030** to naturally urge the indicator mount **1020**, including the indicator shaft **1028**, to move vertically upward along the longitudinal axis C.sub.1. The bottom of the coil spring **1032** may engage or be mounted to the central shaft **1004** at a bottom (not visible) of the central aperture **1016**. The top of the coil spring **1032** may engage a portion the indicator mount **1020**, such as the bottom of the lower extension **1022**. When the indicator mount **1020** is properly mounted to the central shaft **1004** and the indicator assembly **1000** is assembled within the drive housing (FIG. **6**), the coil spring **1032** may be compressed within the central aperture **1016**. When the indicator shaft **1028** is released, as described below, the spring force of the coil spring **1032** will cause the indicator shaft **1028** to move upwards and along the longitudinal axis C.sub.1.
(93) Actuation of the indicator assembly **1000** from the non-activated state, where the indicator shaft **1028** is not visible to the user (operator), to the activated state, where the indicator shaft **1028** is visible to the user, will be dictated by actuation of the first drive input **406***a*. Once the predetermined operational threshold of the surgical tool **200** (FIG. **2**) has been reached and acknowledged, the robotic manipulator of the tool driver may be programmed to actuate the first drive input **406***a* and thereby transition the indicator assembly **1000** to the activated state.
(94) More specifically, the first drive input **406***a* may be rotated, which will correspondingly rotate the central shaft **1004** in the same angular direction. The coil spring **1006** will resist rotation of the first drive input **406***a*, but the spring force of the coil spring **1006** can be overcome with sufficient torque provided by the first drive input **406***a*. As the central shaft **1004** rotates, the indicator mount **1020** remains stationary relative to the receptacle **1014** since the lateral projection **1024** is rotationally secured to a rotationally static portion of the drive housing **208**, as generally described above.
(95) Continuing to overcome the spring force of the coil spring **1006** and thereby rotating the central shaft **1004** will eventually align the release notch **1018** with the release member **1026**. Once the release notch **1018** aligns with the release member **1026**, the release member **1026** may be able

to be received within and extend through the release notch, which allows the spring force of the coil spring **1032** to be released. Releasing the spring force of the coil spring **1032** causes the indicator mount **1020** and, more particularly, the indicator shaft **1028**, to move vertically along the central axis C.sub.1, thus enabling the indicator shaft **1028** to extend through the indicator aperture **502** (FIGS. **5**A-**5**B) of the drive housing **208** (FIGS. **5**A-**5**B and **6**). Extending the indicator shaft **1028** through the indicator aperture **502** effectively transitions the indicator assembly **1000** to the activated state.
(96) Notably, the spring force of the coil spring **1006** may be rated high enough such that a user (technician) that is cleaning or sterilizing the surgical tool **200** (FIG. **2**) will be practically unable to manually overcome the spring force. Consequently, manually and inadvertently actuating the indicator assembly **1000** prematurely will occur in only extremely rare circumstances.
(97) Embodiments disclosed herein include:
(98) A. A surgical tool that includes a drive housing defining an indicator aperture, a drive input rotatably coupled to a bottom of the drive housing, and an indicator assembly arranged within the drive housing and actuatable to provide visual indication that the surgical tool has exhausted its useful life, the indicator assembly including a capstan assembly operatively coupled to the drive input such that rotation of the drive input correspondingly rotates the capstan assembly, an indicator mount secured to the drive housing, and an indicator shaft extending from the indicator mount along a longitudinal axis coaxially aligned with the indicator aperture, wherein the capstan assembly is rotated with the drive input to actuate the indicator assembly between a non-activated state, where the indicator shaft is recessed into the indicator aperture, and an activated state, where the indicator shaft extends out of the drive housing via the indicator aperture to provide the visual indication.
(99) B. A method of operating a surgical tool that includes determining that a useful life of the surgical tool has been exhausted, the surgical tool including a drive housing defining an indicator aperture, a drive input rotatably coupled to a bottom of the drive housing, and an indicator assembly arranged within the drive housing, the indicator assembly including a capstan assembly operatively coupled to the drive input such that rotation of the drive input correspondingly rotates the capstan assembly, an indicator mount secured to the drive housing, and an indicator shaft extending from the indicator mount along a longitudinal axis coaxially aligned with the indicator aperture. The method further including rotating the drive input and thereby rotating the capstan assembly and actuating the indicator assembly between a non-activated state, where the indicator shaft is recessed into the indicator aperture, and an activated state, where the indicator shaft extends out of the drive housing via the indicator aperture, and providing a visual indication with the indicator shaft that the useful life of the surgical tool has been exhausted.
(100) Each of embodiments A and B may have one or more of the following additional elements in any combination: Element 1: wherein the indicator assembly is actuated to the activated state by rotating the capstan assembly with the drive input in a predetermined rotational sequence comprising a combination of clockwise and counter-clockwise rotations. Element 2: wherein the indicator mount is secured to the drive housing such that the indicator mount and the indicator shaft are constrained within the drive housing to movement along the longitudinal axis, the indicator assembly further including a follower extending from the indicator mount and engageable with the capstan assembly. Element 3: wherein the capstan assembly includes a plurality of structural levels axially offset from each other along a central axis of the capstan assembly, each structural level including a planar surface parallel to the planar surface of vertically adjacent structural levels, and one or more ramps providing a vertical transition between the vertically adjacent structural levels, wherein, as the capstan assembly is rotated in the predetermined rotational sequence, the follower progressively and sequentially engages each structural level by engaging and riding up each ramp, and wherein, as the follower transitions to each structural level, the indicator mount and the indicator shaft correspondingly move vertically along the longitudinal axis and toward the indicator

aperture. Element 4: wherein the capstan assembly further includes a pocket defined in a top-most structural level of the plurality of structural levels, and wherein the follower is received within the pocket upon completing the predetermined rotational sequence. Element 5: wherein the indicator assembly further includes a cutout defined in the capstan assembly and extending to a bottom-most structural level of the plurality of structural levels, and a coil spring extending about the indicator shaft and operable to build spring force as the follower progressively and sequentially engages each structural level, wherein, when the capstan assembly is not rotated in the predetermined rotational sequence, the follower falls into the cutout and the coil spring releases the spring force to urge the follower to the bottom-most structural level. Element 6: wherein a bottom of the coil spring engages the indicator mount, and a top of the coil spring engages a static portion of the drive housing. Element 7: wherein at least one of the one or more ramps comprises a living hinge allowing the follower to traverse under the at least one of the one or more ramps when the capstan assembly is rotated in a first angular direction, and allowing the follower to engage and ride up the at least one of the one or more ramps when the capstan assembly is rotated in a second angular direction opposite the first angular direction. Element 8: wherein the indicator mount is rotatably mounted to the drive housing and rotatable about the longitudinal axis, the indicator assembly further including an arm extending from the indicator mount, and a follower extending from an end of the arm, wherein the arm is spring-loaded and urges the follower into engagement with the capstan assembly. Element 9: wherein the capstan assembly includes a plurality of disc keys coaxially aligned with a central axis of the capstan assembly, each disc key defining a slot, wherein, as the capstan assembly is rotated in the predetermined rotational sequence, the slot in each disc key aligns with vertically adjacent slots to form a channel in the capstan assembly, and wherein the follower is biased into the channel with the arm as the indicator mount rotates about the longitudinal axis. Element 10: wherein the indicator shaft is releasably coupled to the indicator mount as the indicator mount rotates about the longitudinal axis, and wherein a coil spring is arranged to urge the indicator shaft vertically along the longitudinal axis when the indicator shaft is released from the indicator mount. Element 11: wherein the capstan assembly further includes a central shaft having opposing lower and upper ends, the lower end extending from the drive input, a torsion spring extending about the central shaft and having a first end operatively coupled to the drive input and a second end secured to the drive housing such that rotation of the central shaft is resisted by the torsion spring, and a receptacle provided at the upper end and defining a central aperture sized to releasably receive a portion of the indicator mount, wherein the indicator assembly is actuated to the activated state by rotating the central shaft with the drive input to overcome spring force of the torsion spring, and thereby release the indicator mount and the indicator shaft from the receptacle. Element 12: wherein the indicator mount provides a lateral projection secured to a rotationally static portion of the drive housing. Element 13: wherein the receptacle defines a release notch and the indicator mount provides a release member alignable with the release notch as the central shaft is rotated to overcome the spring force, and wherein the capstan assembly further includes a coil spring that releases and causes the indicator mount and the indicator shaft to move vertically along the longitudinal axis when the release notch aligns with the release member.

(101) Element 14: wherein rotating the drive input and thereby actuating the indicator assembly between the non-activated state and the activated state comprises rotating the capstan assembly with the drive input in a predetermined rotational sequence comprising a combination of clockwise and counter-clockwise rotations. Element 15: wherein the indicator assembly further includes a follower extending from the indicator mount and engageable with the capstan assembly, and the capstan assembly includes a plurality of structural levels axially offset from each other and each structural level including a planar surface parallel to the planar surface of vertically adjacent structural levels, the method further comprising rotating the capstan assembly in the predetermined rotational sequence with the drive input, progressively and sequentially engaging each structural

level with the follower as the capstan assembly is rotated in the predetermined rotational sequence, and moving the indicator mount and the indicator shaft vertically along the longitudinal axis and toward the indicator aperture as the follower transitions to each structural level. Element 16: wherein the capstan assembly further includes one or more ramps providing a vertical transition between the vertically adjacent structural levels, the method further comprising engaging the follower on the one or more ramps and thereby transitioning the follower between vertically adjacent structural levels of the plurality of structural levels, and moving the indicator mount and the indicator shaft vertically along the longitudinal axis and toward the indicator aperture as the follower engages the one or more ramps. Element 17: wherein the indicator assembly further includes a cutout defined in the capstan assembly and extending to a bottom-most structural level of the plurality of structural levels, and a coil spring extending about the indicator shaft, the method further comprising rotating the capstan assembly in a manner different from the predetermined rotational sequence, encountering the cutout with the follower as the capstan assembly is rotated in the manner different from the predetermined rotational sequence, and returning the follower to the bottom-most structural level as the follower falls into the cutout. Element 18: wherein at least one of the one or more ramps comprises a living hinge, the method further comprising allowing the follower to traverse under the at least one of the one or more ramps when the capstan assembly is rotated in a first angular direction, and allowing the follower to engage and ride up the at least one of the one or more ramps when the capstan assembly is rotated in a second angular direction opposite the first angular direction. Element 19: wherein the indicator assembly further includes an arm extending from the indicator mount, and a follower extending from an end of the arm, and wherein the capstan assembly includes a plurality of disc keys and coaxially aligned with a central axis of the capstan assembly, each disc key defining a slot, the method further comprising rotating the capstan assembly in the predetermined rotational sequence with the drive input, aligning the slot of each disc key with vertically adjacent slots and thereby forming a channel in the capstan assembly, and biasing the follower into the channel with the arm once the channel is formed. Element 20: wherein the indicator shaft is releasably coupled to the indicator mount as the indicator mount rotates about the longitudinal axis, the method further comprising urging the indicator shaft vertically along the longitudinal axis with a coil spring when the indicator shaft is released from the indicator mount. Element 21: wherein the capstan assembly further includes a central shaft extending from the drive input, a torsion spring extending about the central shaft, and a receptacle provided at the upper end and defining a central aperture sized to releasably receive a portion of the indicator mount, the method further comprising resisting rotation of the central shaft with the torsion spring, and operating the drive input to overcome a spring force of the torsion spring and thereby releasing the indicator mount and the indicator shaft from the receptacle. Element 22: wherein the receptacle defines a release notch and the indicator mount provides a release member alignable with the release notch, the method further comprising rotating the central shaft and thereby aligning the release notch with the release member, releasing the indicator mount and the indicator shaft from the receptacle once the release notch aligns with the release member, and moving the indicator mount and the indicator shaft vertically along the longitudinal axis with a coil spring when the release notch aligns with the release member. Element 23: wherein the capstan assembly further includes a drive gear arranged to drive a driven gear positioned within the drive housing, and wherein determining that the useful life of the surgical tool has been exhausted is preceded by rotating the drive input and thereby driving the driven gear with the drive gear, and performing a secondary function of the surgical tool as the driven gear is rotated with the drive gear.

(102) By way of non-limiting example, exemplary combinations applicable to A and B include: Element 1 with Element 2; Element 2 with Element 3; Element 3 with Element 4; Element 3 with Element 5; Element 5 with Element 6; Element 3 with Element 7; Element 1 with Element 8; Element 8 with Element 9; Element 9 with Element 10; Element 11 with Element 12; Element 11

with Element 13; Element 14 with Element 15; Element 15 with Element 16; Element 17 with Element 18; Element 17 with Element 19; Element 15 with Element 19; Element 19 with Element 20; and Element 21 with Element 22.

(103) Therefore, the disclosed systems and methods are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular embodiments disclosed above are illustrative only, as the teachings of the present disclosure may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular illustrative embodiments disclosed above may be altered, combined, or modified and all such variations are considered within the scope of the present disclosure. The systems and methods illustratively disclosed herein may suitably be practiced in the absence of any element that is not specifically disclosed herein and/or any optional element disclosed herein. While compositions and methods are described in terms of “comprising,” “containing,” or “including” various components or steps, the compositions and methods can also “consist essentially of” or “consist of” the various components and steps. All numbers and ranges disclosed above may vary by some amount. Whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range is specifically disclosed. In particular, every range of values (of the form, “from about a to about b,” or, equivalently, “from approximately a to b,” or, equivalently, “from approximately a-b”) disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. Moreover, the indefinite articles “a” or “an,” as used in the claims, are defined herein to mean one or more than one of the elements that it introduces. If there is any conflict in the usages of a word or term in this specification and one or more patent or other documents that may be incorporated herein by reference, the definitions that are consistent with this specification should be adopted.

(104) As used herein, the phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

Claims

1. A surgical tool, comprising: a drive housing defining an indicator aperture; a drive input rotatably coupled to a bottom of the drive housing; and an indicator assembly arranged within the drive housing and actuatable to provide visual indication that the surgical tool has exhausted its useful life, the indicator assembly including: a capstan assembly operatively coupled to the drive input such that rotation of the drive input correspondingly rotates the capstan assembly; an indicator mount secured to the drive housing; and an indicator shaft extending from the indicator mount along a longitudinal axis coaxially aligned with the indicator aperture, wherein the capstan assembly is rotated with the drive input to actuate the indicator assembly between a non-activated state, where the indicator shaft is recessed into the indicator aperture, and an activated state, where the indicator shaft extends out of the drive housing via the indicator aperture to provide the visual indication.

2. The surgical tool of claim 1, wherein the indicator assembly is actuated to the activated state by rotating the capstan assembly with the drive input in a predetermined rotational sequence comprising a combination of clockwise and counter-clockwise rotations.

3. The surgical tool of claim 2, wherein the indicator mount is secured to the drive housing such that the indicator mount and the indicator shaft are constrained within the drive housing to movement along the longitudinal axis, the indicator assembly further including a follower extending from the indicator mount and engageable with the capstan assembly.
4. The surgical tool of claim 3, wherein the capstan assembly includes: a plurality of structural levels axially offset from each other along a central axis of the capstan assembly, each structural level including a planar surface parallel to the planar surface of vertically adjacent structural levels; and one or more ramps providing a vertical transition between the vertically adjacent structural levels, wherein, as the capstan assembly is rotated in the predetermined rotational sequence, the follower progressively and sequentially engages each structural level by engaging and riding up each ramp, and wherein, as the follower transitions to each structural level, the indicator mount and the indicator shaft correspondingly move vertically along the longitudinal axis and toward the indicator aperture.
5. The surgical tool of claim 4, wherein the capstan assembly further includes a pocket defined in a top-most structural level of the plurality of structural levels, and wherein the follower is received within the pocket upon completing the predetermined rotational sequence.
6. The surgical tool of claim 4, wherein the indicator assembly further includes: a cutout defined in the capstan assembly and extending to a bottom-most structural level of the plurality of structural levels; and a coil spring extending about the indicator shaft and operable to build spring force as the follower progressively and sequentially engages each structural level, wherein, when the capstan assembly is not rotated in the predetermined rotational sequence, the follower falls into the cutout and the coil spring releases the spring force to urge the follower to the bottom-most structural level.
7. The surgical tool of claim 6, wherein a bottom of the coil spring engages the indicator mount, and a top of the coil spring engages a static portion of the drive housing.
8. The surgical tool of claim 4, wherein at least one of the one or more ramps comprises a living hinge allowing the follower to traverse under the at least one of the one or more ramps when the capstan assembly is rotated in a first angular direction, and allowing the follower to engage and ride up the at least one of the one or more ramps when the capstan assembly is rotated in a second angular direction opposite the first angular direction.
9. The surgical tool of claim 2, wherein the indicator mount is rotatably mounted to the drive housing and rotatable about the longitudinal axis, the indicator assembly further including: an arm extending from the indicator mount; and a follower extending from an end of the arm, wherein the arm is spring-loaded and urges the follower into engagement with the capstan assembly.
10. The surgical tool of claim 9, wherein the capstan assembly includes a plurality of disc keys coaxially aligned with a central axis of the capstan assembly, each disc key defining a slot, wherein, as the capstan assembly is rotated in the predetermined rotational sequence, the slot in each disc key aligns with vertically adjacent slots to form a channel in the capstan assembly, and wherein the follower is biased into the channel with the arm as the indicator mount rotates about the longitudinal axis.
11. The surgical tool of claim 10, wherein the indicator shaft is releasably coupled to the indicator mount as the indicator mount rotates about the longitudinal axis, and wherein a coil spring is arranged to urge the indicator shaft vertically along the longitudinal axis when the indicator shaft is released from the indicator mount.
12. The surgical tool of claim 1, wherein the capstan assembly further includes: a central shaft having opposing lower and upper ends, the lower end extending from the drive input; a torsion spring extending about the central shaft and having a first end operatively coupled to the drive input and a second end secured to the drive housing such that rotation of the central shaft is resisted by the torsion spring; and a receptacle provided at the upper end and defining a central aperture sized to releasably receive a portion of the indicator mount, wherein the indicator assembly is

actuated to the activated state by rotating the central shaft with the drive input to overcome spring force of the torsion spring, and thereby release the indicator mount and the indicator shaft from the receptacle.

13. The surgical tool of claim 12, wherein the indicator mount provides a lateral projection secured to a rotationally static portion of the drive housing.

14. The surgical tool of claim 12, wherein the receptacle defines a release notch and the indicator mount provides a release member alignable with the release notch as the central shaft is rotated to overcome the spring force, and wherein the capstan assembly further includes a coil spring that releases and causes the indicator mount and the indicator shaft to move vertically along the longitudinal axis when the release notch aligns with the release member.

15. A method of operating a surgical tool, comprising: determining that a useful life of the surgical tool has been exhausted, the surgical tool including a drive housing defining an indicator aperture, a drive input rotatably coupled to a bottom of the drive housing, and an indicator assembly arranged within the drive housing, the indicator assembly including: a capstan assembly operatively coupled to the drive input such that rotation of the drive input correspondingly rotates the capstan assembly; an indicator mount secured to the drive housing; and an indicator shaft extending from the indicator mount along a longitudinal axis coaxially aligned with the indicator aperture; rotating the drive input and thereby rotating the capstan assembly and actuating the indicator assembly between a non-activated state, where the indicator shaft is recessed into the indicator aperture, and an activated state, where the indicator shaft extends out of the drive housing via the indicator aperture; and providing a visual indication with the indicator shaft that the useful life of the surgical tool has been exhausted.

16. The method of claim 15, wherein rotating the drive input and thereby actuating the indicator assembly between the non-activated state and the activated state comprises rotating the capstan assembly with the drive input in a predetermined rotational sequence comprising a combination of clockwise and counter-clockwise rotations.

17. The method of claim 16, wherein the indicator assembly further includes a follower extending from the indicator mount and engageable with the capstan assembly, and the capstan assembly includes a plurality of structural levels axially offset from each other and each structural level including a planar surface parallel to the planar surface of vertically adjacent structural levels, the method further comprising: rotating the capstan assembly in the predetermined rotational sequence with the drive input; progressively and sequentially engaging each structural level with the follower as the capstan assembly is rotated in the predetermined rotational sequence; and moving the indicator mount and the indicator shaft vertically along the longitudinal axis and toward the indicator aperture as the follower transitions to each structural level.

18. The method of claim 17, wherein the capstan assembly further includes one or more ramps providing a vertical transition between the vertically adjacent structural levels, the method further comprising: engaging the follower on the one or more ramps and thereby transitioning the follower between vertically adjacent structural levels of the plurality of structural levels; and moving the indicator mount and the indicator shaft vertically along the longitudinal axis and toward the indicator aperture as the follower engages the one or more ramps.

19. The method of claim 18, wherein the indicator assembly further includes a cutout defined in the capstan assembly and extending to a bottom-most structural level of the plurality of structural levels, and a coil spring extending about the indicator shaft, the method further comprising: rotating the capstan assembly in a manner different from the predetermined rotational sequence; encountering the cutout with the follower as the capstan assembly is rotated in the manner different from the predetermined rotational sequence; and returning the follower to the bottom-most structural level as the follower falls into the cutout.

20. The method of claim 18, wherein at least one of the one or more ramps comprises a living hinge, the method further comprising: allowing the follower to traverse under the at least one of the

one or more ramps when the capstan assembly is rotated in a first angular direction; and allowing the follower to engage and ride up the at least one of the one or more ramps when the capstan assembly is rotated in a second angular direction opposite the first angular direction.

21. The method of claim 16, wherein the indicator assembly further includes an arm extending from the indicator mount, and a follower extending from an end of the arm, and wherein the capstan assembly includes a plurality of disc keys and coaxially aligned with a central axis of the capstan assembly, each disc key defining a slot, the method further comprising: rotating the capstan assembly in the predetermined rotational sequence with the drive input; aligning the slot of each disc key with vertically adjacent slots and thereby forming a channel in the capstan assembly; and biasing the follower into the channel with the arm once the channel is formed.

22. The method of claim 21, wherein the indicator shaft is releasably coupled to the indicator mount as the indicator mount rotates about the longitudinal axis, the method further comprising urging the indicator shaft vertically along the longitudinal axis with a coil spring when the indicator shaft is released from the indicator mount.

23. The method of claim 15, wherein the capstan assembly further includes a central shaft extending from the drive input, a torsion spring extending about the central shaft, and a receptacle provided at the upper end and defining a central aperture sized to releasably receive a portion of the indicator mount, the method further comprising: resisting rotation of the central shaft with the torsion spring; and operating the drive input to overcome a spring force of the torsion spring and thereby releasing the indicator mount and the indicator shaft from the receptacle.

24. The method of claim 23, wherein the receptacle defines a release notch and the indicator mount provides a release member alignable with the release notch, the method further comprising: rotating the central shaft and thereby aligning the release notch with the release member; releasing the indicator mount and the indicator shaft from the receptacle once the release notch aligns with the release member; and moving the indicator mount and the indicator shaft vertically along the longitudinal axis with a coil spring when the release notch aligns with the release member.

25. The method of claim 15, wherein the capstan assembly further includes a drive gear arranged to drive a driven gear positioned within the drive housing, and wherein determining that the useful life of the surgical tool has been exhausted is preceded by: rotating the drive input and thereby driving the driven gear with the drive gear; and performing a secondary function of the surgical tool as the driven gear is rotated with the drive gear.